LATENT-RCT

AI foundation-model representations for confounding control in target trial emulation of cardiovascular RCTs

completed | planned

1 Data

DEVELOPMENT SITE

Yale New Haven Health
index dates 2011–2024

EHR (OMOP) | Raw 12-lead ECG | Echo reports | Death records

EXTERNAL VALIDATION

MIMIC-IV + MIMIC-IV-ECG
Beth Israel Deaconess, Boston
✓ DONE ECG only; 8 trials

UK Biobank
cohort with ECG and cardiac MRI
PLANNED

dbGaP / BioData Catalyst
NHLBI trial and cohort data
ACCORD | WHI | CHS
raw ECG waveforms on BDC
PLANNED

2 Target trials

99 candidate RCTs
feasibility
37 feasible
emulation quality
31 analysed

Quality of the 37 (points-based)

Excellent 3 | Good 11
Moderate 17 | Limited 6 (out)

Analysed 31 by clinical area

AF	11
Diabetes	7
Heart failure	5
Other	4
Hypertension	2
ACS / MI	2

MIMIC-IV set
7 CV trials
PEPTIC (negative control)
in-hospital initiations

3 Representations in the PS

Structured EHR covariates
PS ladder, increasingly rich

PS-Demo	age, sex, index year
PS-CVD5	+ 5 CV diagnoses PRIMARY
PS-CVD10	+ 10 diagnoses
hdPS	+ 200 code features
PS-Clinical	+ vitals, labs, meds (no LVEF; no echo in any PS)

AI-ECG foundation model
raw waveform → BCL encoder → 32 PCs of 256-d embedding

EHR foundation model
code history → CLMBR-T → 64 PCs of 768-d embedding

Placebo: permuted embedding
shuffled between patients

Each PS level: alone, + ECG, + CLMBR-T, + placebo

4 Emulation design

New-user, active-comparator
drug A vs B
look-back ≤365 d | time 0 | follow-up to trial horizon

Propensity-score matching
1:1 nearest neighbour
caliper 0.2 SD of the PS logit

Held-out characteristics
never in any PS
54 non-coded measurements
19 vitals and labs
35 echo measures
✓ leakage-corrected masking

5 Evaluation

Covariate balance PRIMARY
held-out |SMD| after matching; relative reduction

Agreement with RCTs
RCT-DUPLICATE metrics; benchmark-permutation test

Plasmode simulation
withheld confounder (LVEF, NT-proBNP, BMI, eGFR): % bias removed

Robustness
matching, weighting, trial sets, cohort specs, leave-one-out

Counts are numbers of trials or held-out measurements; no patient-level data are shown. ACS, acute coronary syndrome; AF, atrial fibrillation; BCL, biometric contrastive learning; BDC, BioData Catalyst; CV, cardiovascular; hdPS, high-dimensional PS; LVEF, left ventricular ejection fraction; MI, myocardial infarction; PC, principal component; PS, propensity score; SMD, standardized mean difference.